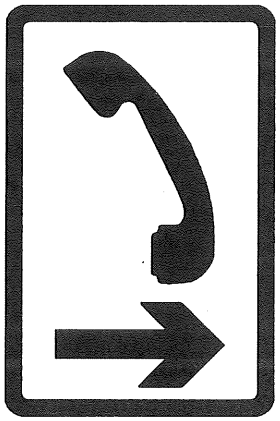
INTRODUCTION TO SIGNALLING

15

TELECOMMUNICATIONS



A. Wilding

INTRODUCTION

The aim of this Section is to provide an appreciation of the telecommunications systems found on today's railway system in Britain. These systems include line side telephones, the Signal box telephone concentrator, transmission systems, SSI long distance data links, radio systems and Customer Information Systems. The requirements of these systems and their applications will be discussed.

Telecommunications has been at the heart of railway signalling from the very beginning. It was introduced in the form of a telegraph between stations to give advanced warning of trains. The railway was given a licence to provide telecoms services in the late 1800s, well before the Post Office received its licence.

As signalling for trains developed, so did the associated operational telecoms, which in turn led to the formation of the engineering organisation of Signalling & Telecommunications within British Rail. The process of privatisation then split this organisation into signalling and telecommunications, its two main constituent parts, thus segregating the two core businesses.

TELECOMMUNICATION SYSTEMS

The railway telecommunications network is based on a number of systems working together to transfer voice and data between operational and business sites. This section will cover the systems required to support voice services, SSI data transmission and customer information

Voice Communication Systems

The UK signalling system is designed to fall back to a safe state of all signals red when a failure occurs. Under these conditions, voice communication is vital for the operation of the railway. For this reason there are many telephone types located along the railway line allowing drivers and operational staff to communicate with the signaller, controller or emergency service. The following paragraphs cover telephone handsets, routing of telephone circuits and the telephone concentrator equipment.

Lineside Telephones

Lineside telephones are the most familiar type of railway telecoms. They are designed for attachment to posts, buildings or other lineside structures. The equipment is weather proof to IP65 which is adequate for normal weather conditions. The telephone case has various labels attached, for example telephone type, telephone location, 6 figure grid reference, signal box to which it is connected, telephone number and other useful numbers to dial in an emergency.

The standard telephone is vandal resistant but where there is a particular problem vulnerability can be reduced. The most common form of vandalism is for the handset to be removed, fitting a metal armoured handset cord or a flush metal plate type telephone with a push button generally alleviates the problem.

Signal Post Telephones

The *Signal Post Telephone*, or SPT, is the most common type and provides a direct "hot-line" between a lineside signal and the signal box. SPTs are provided on or adjacent to all signals that are capable of displaying a red aspect. The rules state that if a driver is stopped at a red signal for more than a minute or so, the signaller must be contacted. This is done by the driver leaving the train cab and using the SPT. Therefore, if the signaller wants to speak with a driver, a signal can be set to red causing the train to stop and the driver to telephone in. The driver can then be informed of urgent information regarding failures, accidents or emergencies. Operational faulting and maintenance staff also use the SPT in the course of their duties, for example, when requesting a possession of the line.

As SPTs are the fall back system during signal failure, the probability of more than one SPT failing in a single location must be minimised. This is achieved by arranging the telephone circuits in the *main bearer cable* such that successive or adjacent telephones are routed back to the signal box in alternate directions. The diversely routed circuits are connected to an apparatus room in the opposite direction from the signal box. At the apparatus room the circuits are connected to a telecoms transmission system to carry the circuits back to the signal box in a different cable.

The telephone is connected to the main bearer cable by a 2 pair tail cable. The bearer cable consists of approximately 100 pairs and carries all the circuits back to the telephone concentrator at the signal box directly or via the diverse route.

Level Crossing Telephones

Level Crossings also have telephones and these are also direct "hot-lines" to the signaller. They are located at either side of the barriers and on the approach roads. They allow people using the crossing to contact the signaller if for example, they have an abnormal load or a broken down vehicle foul of the line. If there is a signal failure a Hand Signaller will be located at the crossing to flag trains through when instructed to do so by the signaller. A telephone is also located adjacent to the barrier lifting equipment in case the barrier fails and manual operation is required. The specific location of the telephones is in accordance with the site layout plan. The telephones are all connected together and presented to the signaller as one line.

Some crossings are fitted with a *Public Emergency Telephone System (PETS).* This intelligent system provides control and indication facilities for up to 4 telephone circuits at a single location and is operated over a single pair of wires.

Alarms and indications include:

- Call in progress.
- PETS equipment and link OK.
- Communications fail from the outstation.
- Remote equipment failure or damage.
- Hand set or multiple hand set loss.
- PETS module failure.

Signal boxes have Direct Line telephones that are "hot-lines" connected to adjacent signal boxes, Electrical Control Rooms (ECR) and station staff.

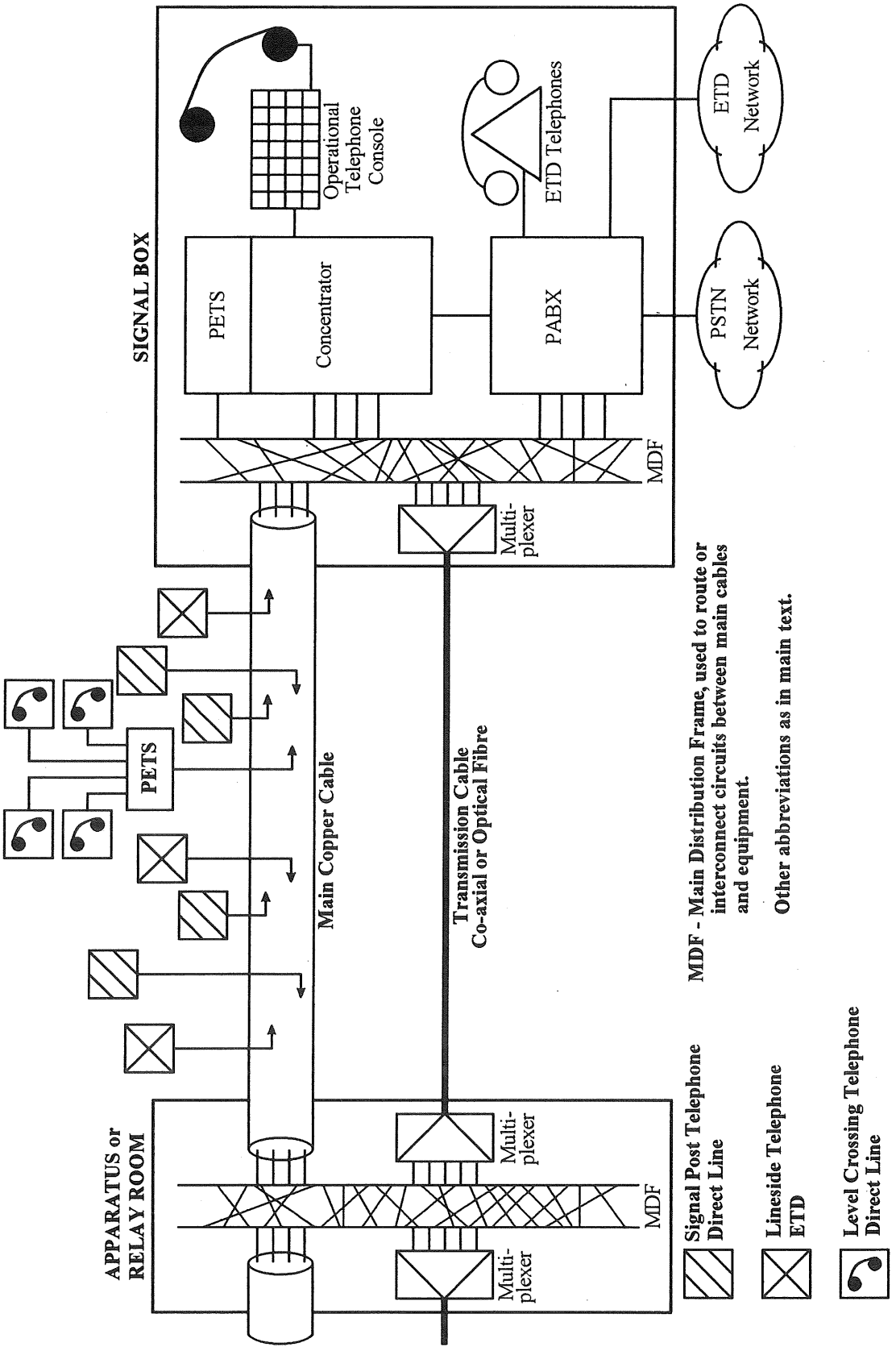
Other Operational Telephones

Other operational telephones are connected to the *Extension Trunk Dialling (ETD)* network but presented to the signaller via the signal box telephone concentrator. The ETD network is the railways own internal telephone network and has similar functionality to the national BT network. These telephones have a number pad allowing the user to dial any number on the railway telephone network. These telephones are commonly called "Auto"s as they are connected to an Automatic Telephone Exchange. ETD telephones are found at sets of points, access points, platforms, substations, ground frames and patrolman lockout locations.

Maintenance Plug Points were designed to allow technicians to use plug-in telephones to communicate along the line between apparatus cases. The original system used a single pair of wires running between the relay room and adjacent apparatus cases. The technicians plugged in a portable battery powered telephone when they needed to speak to each other. This system was developed into a two pair system where each pair was allocated an individual telephone number. Each user connects to a different pair and then dials the appropriate number to speak to the other party. This system also allows other parties on the ETD network to be contacted and can reduce the number of lineside ETD telephones required.

Summary

A summary of the various systems described is shown in the diagram overleaf.

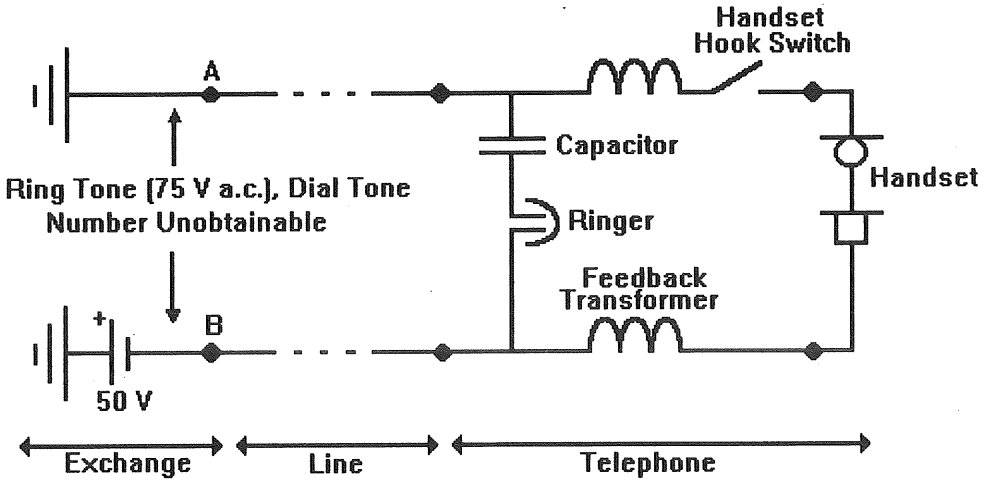

SIGNAL BOX
PETS
Concentrator
Operational Telephone Console
ETD Telephones
PABX
ETD Network
PSTN Network
MDF
Multi-plexer
Main Copper Cable
Transmission Cable
Co-axial or Optical Fibre
APPARATUS or RELAY ROOM
Multi-plexer
MDF
Multi-plexer
MDF - Main Distribution Frame, used to route or interconnect circuits between main cables and equipment.
Other abbreviations as in main text.
Signal Post Telephone Direct Line
Lineside Telephone ETD
Level Crossing Telephone Direct Line

Telephone Operation

Telephones operate using a single pair of wires and can be powered from the 50 Volt supply at the telephone exchange. This method of operation is known as *Central Battery (CB)* and is effective up to approximately 10 km. When a user lifts the handset to make a call the handset hook switch closes, forming a current loop. When the exchange detects this current, dial tone is generated by the exchange and a call can be made.

For operation over slightly longer distances an additional battery is placed inside the telephone, this is known as *Local Battery (LB)* operation. The LB telephones are sometimes known as 'Magneto's' as early examples were fitted with a wind up generator rather than a battery.

Long distance circuits require a telecoms transmission system that can regenerate the speech signal. In this case the telephone and the exchange are no longer physically connected so an interface is required, this is known as an *E & M interface*. When a user lifts the telephone handset, dial tone is requested from the distant telephone exchange by an earth potential being applied to the M (Mouth) wire. When this is detected on the E (Ear) wire at the exchange, dial tone is generated and the call connected. This E & M signalling is called Direct Current 5 or DC5.



Telephone Concentrator Systems

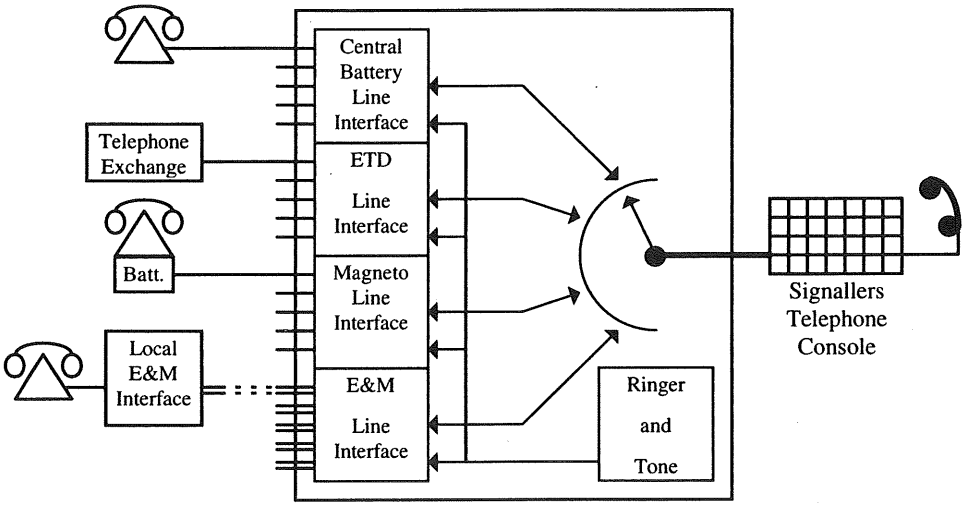
A *telephone concentrator* is a simple telephone exchange that allows a number of lineside telephones to be connected to a single console at the signal box. Concentrators are designed to the requirements of standards BR 1655 and BR 1602.

BR 1655 states:

- Only one call shall be capable of being established between an operator and a remote party at any one time using hardware and software interlocks.
- No interruption or overhearing of the call should be possible under any circumstances.
- The unique identity of each circuit shall be presented to the signaller when an incoming call is received.
- The signaller shall be aware which circuit he is communicating with at all times.

BR 1602 is an augmentation of this for large concentrator systems, allowing up to 11 desks and a total of 1024 lines.

Signal Box Telephone Concentrator



The concentrator has a connection with the P*ublic Switched Telephone Network (PSTN)* allowing the signaller to make calls outside the ETD network. As a result concentrators must have BABT (British Approvals Board for Telecommunications) approval. If the equipment is integrated with other systems such as radio, additional approvals may be required from the RCA (Radio Communications Agency) and OFTEL.

Signaller's can only make and receive calls with SPTs within their area of control and no provision is made for transferring calls or interconnecting SPTs with each other. The signaller can swap between calls without loosing the circuit by parking a call. This is achieved by simply selecting another line, the parked caller then hears a ringing tone and the associated key flashes at a distinctively fast rate on the signal box console. The call can be re-established by pressing the fast flashing key again.

Each telephone has its own individual line on the concentrator to ensure the call is secure and cannot be overheard. The signaller is only able to speak to one caller at a time which is particularly important if the signaller is giving a driver authority to pass a signal at danger or a possession to a member of operational staff.

A voice and data recording system is provided to record all incoming and out going calls from the signallers console. If there is an incident or accusation regarding an authorisation or conversation, Railtrack are responsible for recovering the tapes when they are required as evidence.

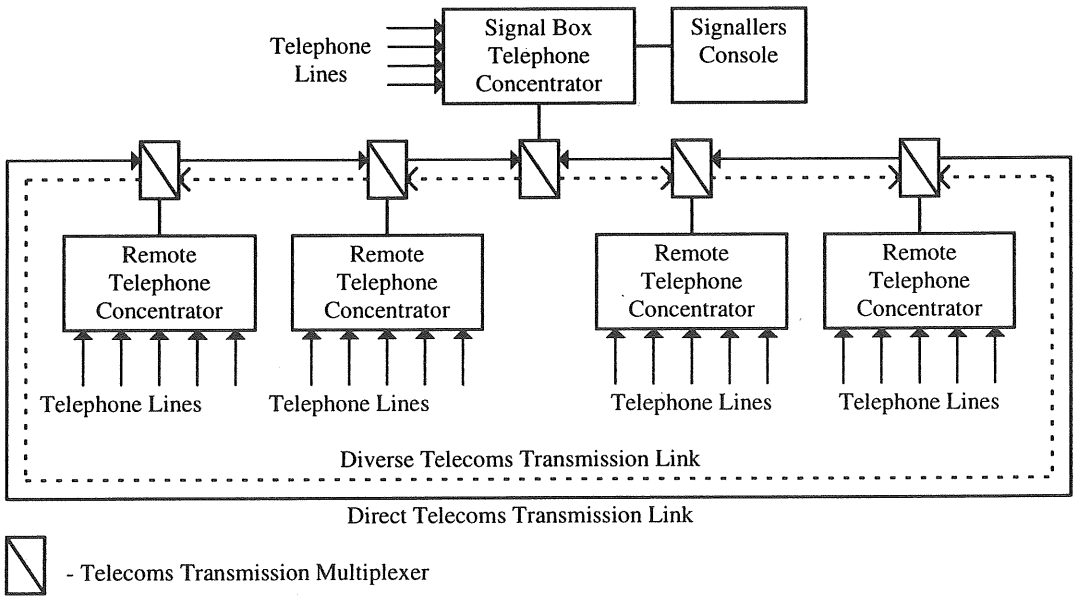
The signallers handset has a Press To Talk (PTT) button that needs to be held down when speaking. This minimises the risk of a driver or trackside worker hearing an instruction directed to someone else. The handsets also have long curly cords allowing the signaller to move along the panel whilst speaking to the person trackside. This is necessary when routes are set or points need to be operated during maintenance.

Emergency calls and calls from Automatic Half Barrier (AHBCs) level crossings are a high priority for the signaller. To ensure the signaller is aware of these calls they are identified by a unique and arresting sound from the concentrator console.

Telephones that are in an area of limited clearance are restricted to out going calls only. If the signaller tries to call one of these telephones the call will not be established, removing the possibility of someone being put in danger by answering a lineside telephone that is not in a position of safety.

Networked Telephone Concentrator Systems

To reduce the number and length of telecoms transmission circuits, a strategy of remotely locating a number of small concentrators in unmanned locations can be implemented. The additional concentrators then combine local telephone circuits before onward transmission to the Signal box, therefore using just one expensive long distance circuit. This level of networking is achieved due to the very low usage of lineside telephones. The benefits are reduced annual line rental charges, reduced maintenance costs to Railtrack and if diverse routing of calls is employed, improved availability of circuits provided.



PABX Solution for the Replacement of Concentrators

The Signal Box telephone concentrator has traditionally been perceived as an expensive, specialised, proprietary piece of equipment available from a limited number of suppliers. It has been found that with minor software changes a commercial *Private Automatic Branch Exchange (PABX)* can provide the required functionality, so increasing the number of possible suppliers and maintainers. This development has led to seven Ericsson PABXs to be installed as trial sites.

The implementation of a PABX solution removes the need for low usage dedicated infrastructure as the system can use inexpensive standard telecoms links. These links are

known as *Digital Private Network Signalling System (DPNSS)* and have standard interfaces so can then be procured competitively from a number of operators.

Voice and data carried over these links can be dynamically re-routed when a failure occurs. Monitoring facilities allow failures to be identified and responded to quickly.

SPT circuits can be programmed into the PABX as "hot lines" and thus be established on demand, re-routing occurring automatically under fault conditions possibly using the public network on a pay as you use basis.

SSI COMMUNICATION LINKS

SSI data is transmitted over short distances using a base band twisted copper pair cable via Data Link Modules (DLMs). There are two links from the Signal Box to the lineside (A and B) improving the robustness of the system by providing diverse routing. Track side Functional Modules (TFMs) are connected to this data link to drive the lineside equipment which include signals and points. The length of the link can be up to 10 km using the copper cable, but distances up to 40 km can be achieved with the use of repeaters.

This 40 km can be remotely located by the use of a telecoms transmission system, effectively extending the area of control to approximately 500 km. A Long Distance Terminal (LDT) provides the interface between the SSI and telecoms transmission system, where each link (A and B) uses a single 64 kbit channel. The LDT encodes additional security data into the message as well as the signal box identity number. The data interface is at 20 kbit/s with an electrical interface defined by the International Telecommunications Union (ITU) recommendation G.703. The A and B messages from the interlocking have timing restrictions that must be acknowledged by the telecoms service provider, these are:

- The A and B messages must arrive at the TFM within 3.6 ms of each other otherwise the message will be discarded.
- When a transmission system is used for the A and B paths of the long line link, and the track side LDT is directly connected to the TFMs, the propagation delay from the interlocking LDT to the track side LDT must not exceed 4.69 ms.
- Where the trackside LDT is connected to a base band data link of up to 40 km, the data link propagation delay must be taken into account. The time for a message to pass from the trackside LDT to the most remote TFM at a maximum of 40 km away, is 0.35 ms. This means the transmission system delay must not exceed 4.34 ms.

As the LDT does not have BABT approval, a telecoms multiplexer normally provides an approved interface.

TRANSMISSION SYSTEM COMMUNICATION LINKS

Transmission systems are used to carry speech and data over long distances in an analogue or digital format using copper or optical fibre cables. The transmission systems use multiplexing to combined many signals into one thus ensuring efficient use of the transmission medium.

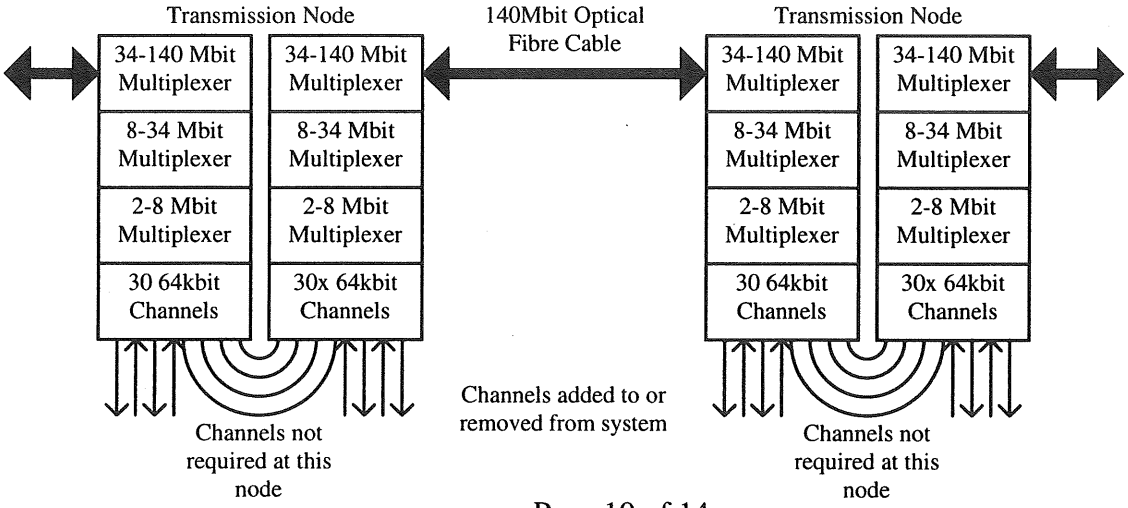
Older systems were analogue and based on *Frequency Division Multiplexing (FDM).* Each speech or data channel was allocated a specific frequency band within the 4 MHz of total bandwidth. Each channel had a bandwidth of 4 kHz enabling 960 channels to be carried within the 4 MHz system.

Current systems use a digital format because of its greater noise immunity and the ability to increase the transmission distances by regenerating rather than amplifying the signal. Channels are allocated a time slot rather than a frequency band by using *Time Division Multiplexing (TDM).*

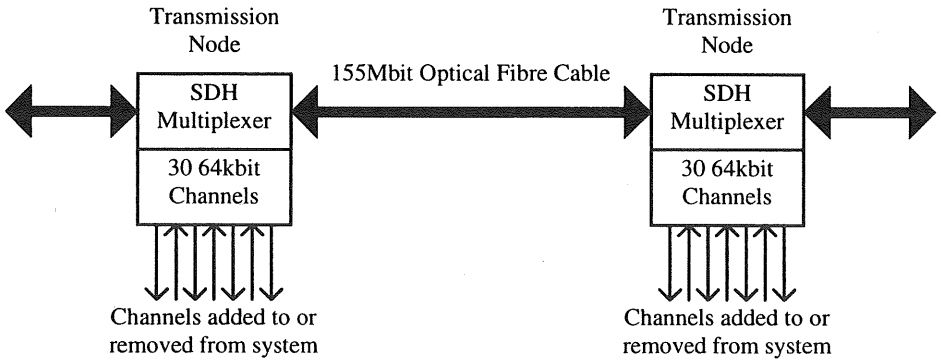
To convert speech or data into a digital format the signal needs to be *sampled.* The samples are then assigned an 8 bit binary value which is the code that is transmitted. This process is called *Pulse Code Modulation (PCM)* and the systems are commonly referred to as a 30 channel or 2 Mbit systems.

When coding speech, Shannon's law states that for the signal to be reproduced intelligibly and to an acceptable quality, the original signal needs to be sampled at twice its highest frequency component. As speech frequencies are typically up to 4000 Hz, sampling is at 8000 Hz. These samples are then allocated the 8 bit binary code. Therefore, the bandwidth of each PCM system speech channel is 64 kbits (8000 Hz by 8 bits). Hence a 2.048 Mbit system can carry 30 channels of speech. There are also 2 channels allocated for system synchronisation and system signalling.

This 2 Mbit system is called first or primary order, these can be multiplexed together to a maximum fourth order system at 140 Mbit/s. The process of combining or multiplexing these systems together is very hardware intensive and the multiplexers are expensive. As speech or data can only be accessed at the 2 Mbit/s level the complete system needs to be de-multiplexed when access to a channel is required. The complete system is then multiplexed up again before onward transmission. This system is called *Plesiochronous Digital Hierarchy (PDH).* Plesiochronous means that the 2 Mbit tributary signals obtain their timing from different clocks operating within a specific tolerance of each other.



Drop and Insert systems were developed to reduce the need for multiplexing hardware at each point that access is required to a channel. This architecture is called *Synchronous Digital Hierarchy (SDH).* SDH uses a nominated master clock for all drop and insert nodes to lock onto. This enables groups or individual channels to be extracted or inserted at any intermediate location along the route. Any channels that are connected through at a node remain coded and in digital format. Channels that are to be extracted or inserted at a node are allocated to a low frequency channel interface unit. Channel units still carry voice or 64 kbit/s of data. The bit rates are slightly higher than PDH at 155 Mbit, 622 Mbit and 2488 Mbit. An SDH network is normally arranged in ring formation that provides a highly flexible, resilient and cost effective solution. A user friendly graphical interface for network management is also used.



CUSTOMER INFORMATION SYSTEMS

Customer information systems provide passengers with information regarding the arrival and departure of trains. These systems include public address, recorded announcements and visual information displayed on screens or boards.

A local Public Address (PA) system allows staff to broadcast messages around the local station by selecting the set of speakers most appropriate to the announcement. Speakers are divided by platform and long platforms have the speakers divided into short and long operation. Short operation reduces the nuisance to neighbours of broadcasts at night. Long line public address systems allow the announcer to broadcast to unmanned stations along the line. These announcements can be automated with recorded speech being broadcast. These systems are often interfaced to the signalling Train Describer allowing the announcements to coincide with the approach of a train.

As with public address system, the visual information system driving the station monitors and display boards is normally connected to the signalling Train Describer to enable the information to be updated automatically. These systems can also be used to operate in a long line format to adjacent stations.

Digital clocks on platforms derive their time signal from the National Physics Laboratory at Rugby. If the signal is lost the clocks will free run, this is often not noticed until British summer times begins or ends and the incorrect time is displayed.

LOCAL RADIO, THE NRN AND CAB SECURE RADIO

The railway has 4 types of radio in operation, *local UHF*, the *National Radio Network (NRN)*, the *Overlay Radio Network (ORN)* and *Cab Secure Radio (CSR).*

Local radio consists of a UHF base station, antenna and controller station. It provides coverage to hand portable radios in a local area, typically a station, depot or engineering work site. This system is then used by staff to co-ordinate and disseminate information.

The ORN radio network covers the majority of railway lines in the UK and was originally developed to provide communication between trains, signal boxes and the ETD network. The NRN uses the same frequency band and fills in gaps in the ORN providing national coverage. The NRN is also used for making calls on the ETD network. All signal boxes have a "Sitronix" telephone to access the radio networks. Unfortunately, if the signaller wants to call a driver, the train location with respect to the radio network and the number of the radio needs to be known. As the country is divided up into approximately 20 individual frequency zones, calling a moving train is difficult. Each of these zones is capable of supporting 16 calls, and is mainly used by operational staff.

CSR was developed to provide SPT functionality using a cab based radio. CSR allows secure communication to and from drivers by using directional antennas focused along the railway line. Train movements are monitored by the CSR database via inputs from the signalling Train Describer. This means any train fitted with CSR can be called at the touch of a button. Features available to the signaller include:

- selective calling of any driver of a CSR equipped train in the area of control.
- general calling to the drivers of all trains in the area of control.
- broadcasting by the signaller over the train's PA system.
- sending a "Stop" message to a driver's display panel.
- connection of a driver to the ETD network if required.

Features available to the driver include:

- request a call to the signaller.
- make an emergency call even if the system is busy with other calls.

The connection to the signalling Train Describer enables the system to automatically change a train's allocated channel as it moves between signaller's areas. Therefore, alterations to the Train Describer database may mean the CSR database also has to be updated. If coverage is poor manual changeover may be required by the driver as indicated by the lineside channel change board.

With CSR there is no need for the driver to leave the train during a journey. The guard's role becomes redundant if the driver can also ensure passengers are safely on or off the train before departure. *Driver Only Operation - Passenger (DOO-P)* CCTV was introduced on station platforms for this purpose as it views the whole train and provides the images to the driver via monitors mounted on the platform. The driver can then view all the train doors whilst the train is standing in the platform and passenger safety is safe guarded. On straight platforms large heated mirrors adequately replace the monitors. There are also trials underway where the pictures are beamed directly into the cab of the train via an infra red data link. They are displayed on LCD screens inside the cab reducing maintenance and faulting on platform equipment.

TRANSMISSION BASED SIGNALLING

As the Trans-European Network (TEN) of high speed lines is developed the EC requires a standardised *European Train Control System (ETCS).* The radio system will be implemented rapidly once the land based equipment and transmission network is established, with the opportunity to recoup some costs by the provision of voice communication services.

The new *Train Control System (TCS)* is being developed to replace existing conventional signalling. The system will incorporate cab signalling which conforms to the *European Rail Traffic Management System (ERTMS).*

TCS will use the GSM radio network for constant communication from a controlling system to the moving train. The frequencies have been allocated for this system and are known as *GSM-Railway (GSM-R).* The system is being developed on the West Coast Main Line and will support a variable moving block signalling system with train speeds up to 250 km/h.

A second radio based solution was considered for TCS based on *the Trans-European Trunked Radio (TETRA)* network. TETRA was designed for data transmission and lacks the concerns associated with maximum permissible time for data transfer and call set up surrounding GSM-R. TETRA will be used by people such as the Home Office, Police and Fire Brigade and suppliers are already producing handsets and base stations. It is expected that GSM-R will be available by 2005.